

Mineral Insulated Metal Sheathed Thermocouple (TC)

A thermocouple is a device consisting of two different conductors, usually metal alloys. When two dissimilar metals are joined together, a hot junction is formed, and the other end will be cold. When these two ends are subjected to different temperatures, an EMF (electromotive force) will be induced inside the conductor which is measured in millivolts(mv). However, if both junctions are at the same temperature, no EMF is generated.

There are various type of thermocouples of which K, J and S are most commonly used. The S type thermocouple is used for higher temperature. The maximum temperature it can withstand depends on the diameter of the wire, the environment and expected durability.



Special Features

Compliance to IEC 60584

Temperature range -200 to 1250 °C

Mineral insulated construction with K type thermocouple

Available in a variety of sheath materials

High sheath ductility and dielectric strength of conductors

Quick response time

Industrial Applications

Chemical and fertiliser industries

Petrochemical industry

Thermal, nuclear and hydroelectric power stations

Pharmaceutical industry

Food processing industry

Water treatment plants

Iron and steel industry

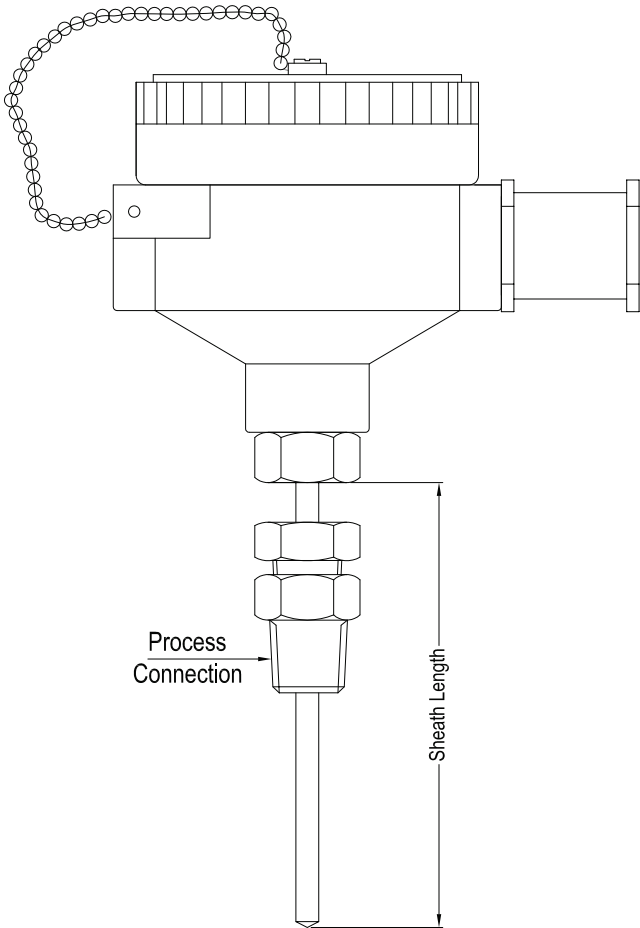
Rubber molding / processing plants

Sugar industry

Refineries

Standard Product Specifications	
Specifications	Description
Reference standard	IEC-60584
Model	Simplex
Type	K-type
Wire configuration	2- wire
Connection	1/2" NPT(M)
Sheath material	SS 316
Sheath type	Mineral insulated measuring insert
Sheath diameter	6 mm
Sheath length	200 mm
Accuracy	Class 1 as per IEC -60584
Terminal head	Screwed type die cast aluminium weatherproof – IP 65
Head extension type	Adjustable connection
Cable gland	3/4" ET single compression gland
Cable gland MOC	Brass nickle plated with rubber
Ambient temperature	-25°C to 65°C
Temperature range	-200 to +1250°C

MI Cable Thermocouple Assembly



Ordering Information

Model	
Thermocouple(duplex)	TD
Thermocouple(simplesx)	TS

Type	
K-type (range -200 to +1250°C)	K
J-type (range 0 to +750°C)	J
S-type (range 0 to +1450°C)	S

Wire Configuration	
2 wire	2
4 wire	4

Head Extension Type	
Adjustable connection	A
Nipple connection	B
Nipple union connection	D
Fixed connection (standard)	F
Flanged connection	E

Connection Size	
M18 X 1P	0
M20 X 1.5P	1
1/4"	A
3/8"	B
1/2"	C
3/4"	D
Not required	N

Connection Type	
NPT(M)	A
BSP(M)	B
BSPT(M)	C
Metric thread(M)	M
#150-RF	4
#300-RF	5
Without connection	N

Cable Gland	
1/2" NPT, double compression(1 No)	A
3/4" ET double compression(1 No)	C
3/4" ET, Single compression(1 No) (standard)	D
1/2" NPT, double compression(2 No)	E
3/4" ET, double compression(2 No)	G
3/4" ET, single compression(2 No)	H

Configuration Generation Code

Sheath Diameter
6 mm (standard)
8 mm
10 mm
3 mm
4.5 mm

Sheath Length
As per requirement

Sheath - Material of Construction
SS 316 (standard)
Inconel 600
Inconel 625
SS 321
SS 446
SS 310

Head Extension - Material of Construction
SS 316 (standard)
SS 304
SS 316L

Cable Gland - Material of Construction
Brass nickle plated with rubber

Optional Features	
Accuracy class 2 as per IEC 60584	
Head protection	IP 67 protected head
	Ex. proof GR II C
	Flame proof GR II A
	Flame proof GR II B
	Flame proof GR II A, II B
Lead wire extension	PTFE braided
	SS braided
Lead wire length	1 mtr to 15 mtrs
Flange type	Blind
	Slip on
	Weld neck
Flange Finish	125 AARH
	63 AARH
	250 AARH

Tests and Certificates	
Five point calibration certificate	
Material test certificate	
Third party inspection	
Special requirement – please specify	

Ordering example: TSK2FNND-XXXX (with standard selection)



Forbes Marshall
Krohne Marshall
Forbes Marshall Arca
Codel International
Forbes Solar
Forbes Vyncke
Forbes Marshall Steam Systems

Plot No. A-19/2 & T-4/2, I.D.A
Nacharam, Hyderabad - 500 076
Tel : +91-40-27179223, 27171909,
27170732
Fax : +91-40-27173235

Email : sales-gauges@forbesmarshall.com, ccmidc@forbesmarshall.com

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B-85, Phase II, Chakan Indl Area
Sawardari, Chakan, Tal. Khed
Dist. Pune - 410 501. INDIA
Tel : 91(0)2135-393400

A-34/35, MIDC H Block
Pimpri, Pune - 411 018. INDIA.
Tel : 91(0)20-27442020, 39851199
Fax : 91(0)20-27442040

CIN No.: U28996PN1985PTC037806

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